

- 2 a** Distance is the actual path travelled, **B1**
 displacement is the straight line distance between start and finish points (in that direction)/ minimum distance **B1**

- 2 b** *Taking downwards as positive*

$$s = ut + \frac{1}{2} at^2$$

$$30 = 0.5 (9.81) t^2$$

C1

$$t = 2.47 \text{ s}$$

A1

- c** Taking downwards as positive

Vertical component

$$v_y^2 = u_y^2 + 2a_y s_y$$

$$v_y^2 = 0 + 2(9.81)(30)$$

$$v_y = 24.26$$

C1

$$\text{resultant velocity : } v^2 = (24.26)^2 + (20)^2$$

$$v = 31.4 \text{ m s}^{-1}$$

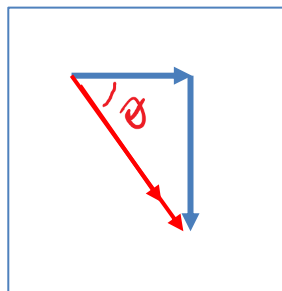
A1

$$\text{Direction : } \tan \theta = \frac{24.26}{20}, \theta = 50.5^\circ$$

Describe as 50.5° below positive x-axis

C1

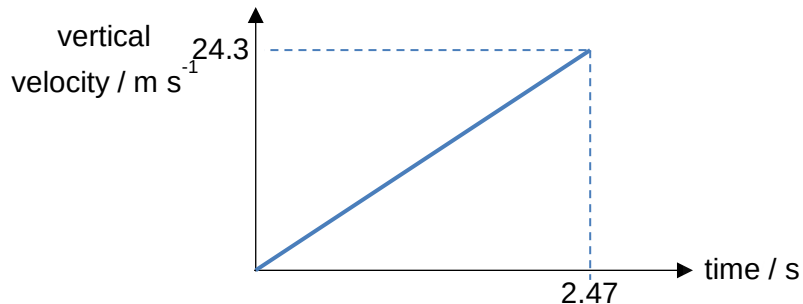
Alternative methods to find vertical component of speed



using $v = u + at$ (allow e.c.f)

initial (potential energy + kinetic energy) = final kinetic energy

d



B1

B1

[1] straight line passing through origin with constant gradient

[1] labelling of values